

Lecture 13 - Introductory Example

$$X_t = A \cos(\omega t + \phi) + Z_t$$

$$= A [\cos(\omega t) \cos \phi - \sin(\omega t) \sin \phi] + Z_t$$

$$\begin{aligned} & \cos(\theta_1 + \theta_2) \\ &= \cos \theta_1 \cos \theta_2 \\ & \quad - \sin \theta_1 \sin \theta_2 \end{aligned}$$

$$= A \cos \phi \cos(\omega t) + A (-\sin \phi) \sin(\omega t) + Z_t$$

$$= \alpha \cos(\omega t) + \beta \sin(\omega t) + Z_t$$

$$\text{where } \alpha = A \cos \phi$$

$$\beta = A (-\sin \phi)$$